

The multiplier algebra of the coefficient-weighted Bergman scale

A literature-implied full answer to the question in arXiv:2105.07760

Autonomous literature and proof audit, lane 18

11 August 2026

Status. Full literature-implied answer at the level of an exact function-theoretic criterion. Nothing here is claimed as new: the ingredients predate the source paper.

1 The source question

Gallardo-Gutiérrez and Partington [1] define, for $\alpha \in \mathbb{R}$,

$$\mathcal{A}_\alpha = \left\{ f(z) = \sum_{n \geq 0} a_n z^n : \|f\|_\alpha^2 := \sum_{n \geq 0} |a_n|^2 (n+1)^\alpha < \infty \right\},$$

and let $\mathcal{M}_\alpha$ be the analytic functions g for which $f \mapsto gf$ is bounded on $\mathcal{A}_\alpha$. On source PDF page 3 they write:

For other instances of α this is no longer the case, and it is an open question to identify them for any α .

Here “the case” refers to the equality with H^∞ at $\alpha = -1, 0$. The literature already supplies a complete phase diagram, although the middle range is naturally described by a Carleson condition rather than by a simpler named algebra.

2 Complete phase diagram

A positive measure μ on $\mathbb{D}$ is called a Carleson measure for $\mathcal{A}_\alpha$ if

$$\int_{\mathbb{D}} |f|^2 d\mu \leq C \|f\|_\alpha^2 \quad (f \in \mathcal{A}_\alpha).$$

Theorem 1. *For every real α , the multiplier algebra of $\mathcal{A}_\alpha$ is as follows:*

$$\begin{aligned} \alpha \leq 0 & : \mathcal{M}_\alpha = H^\infty, \\ 0 < \alpha \leq 1 & : \mathcal{M}_\alpha = \left\{ g \in H^\infty : |g'(z)|^2 (1 - |z|^2)^{1-\alpha} dA(z) \text{ is Carleson for } \mathcal{A}_\alpha \right\}, \\ \alpha > 1 & : \mathcal{M}_\alpha = \mathcal{A}_\alpha. \end{aligned}$$

The identities are set identities, and the corresponding natural norms are equivalent.

The first and third regimes occur already in Taylor’s one-variable multiplier theory [2]. Jupiter and Redett [5], whose paper explicitly extends Taylor’s results, record the stronger polydisc versions as Proposition 3.9 and Theorem 3.10 on physical PDF page 13. The middle regime is the classical Kerman–Sawyer criterion [4], quoted explicitly by Bao, Lou, Qian, and Wulan [6, p. 1722; physical PDF p. 23]. The classical Dirichlet endpoint $\alpha = 1$ is also covered by Stegenga’s multiplier theorem [3].

3 Parameter translation and direct verification

For $0 < \alpha \leq 1$, beta-function asymptotics give the norm equivalence

$$\|f\|_\alpha^2 \asymp |f(0)|^2 + \int_{\mathbb{D}} |f'(z)|^2 (1 - |z|^2)^{1-\alpha} dA(z). \quad (1)$$

Thus the source space is the standard weighted Dirichlet space $\mathcal{D}_K$ with $K(t) = t^{1-\alpha}$, up to an equivalent norm. Since $0 < \alpha \leq 1$, this K is increasing and concave and $t/K(t) = t^\alpha \rightarrow 0$. The Kerman–Sawyer theorem therefore applies. Replacing $1 - |z|$ by $1 - |z|^2$ changes the weight only by fixed constants.

For completeness, the multiplier criterion itself follows immediately from (1). If M_g is bounded, then the reproducing-kernel identity $M_g^* k_z = \overline{g(z)} k_z$ gives $g \in H^\infty$. Moreover,

$$fg' = (fg)' - gf'$$

shows that $|g'|^2 (1 - |z|^2)^{1-\alpha} dA$ is Carleson. Conversely, if g is bounded and that measure is Carleson, then

$$(fg)' = fg' + gf'$$

and (1) give $\|fg\|_\alpha \lesssim \|f\|_\alpha$.

The endpoint regimes are equally direct. For $\alpha < 0$, $\mathcal{A}_\alpha$ has an equivalent radial weighted-Bergman norm, so $H^\infty \subseteq \mathcal{M}_\alpha$; the kernel argument gives the reverse inclusion. The case $\alpha = 0$ is H^2 . If $\alpha > 1$, weighted Cauchy–Schwarz gives ℓ^1 summability of the Taylor coefficients. Together with

$$(m + n + 1)^{\alpha/2} \lesssim (m + 1)^{\alpha/2} + (n + 1)^{\alpha/2},$$

Young’s convolution inequality proves that $\mathcal{A}_\alpha$ is a Banach algebra. Hence $\mathcal{A}_\alpha \subseteq \mathcal{M}_\alpha$, while the reverse inclusion follows by applying a multiplier to 1.

4 Provenance verdict

The three regimes exhaust $\mathbb{R}$, so the source sentence has a full answer at the criterion level. The status is nevertheless *literature-implied*, not “later paper explicitly answered”: the cited theorems predate arXiv:2105.07760 and require the norm/parameter identification above. This packet should not be counted as an original solve.

References

- [1] E. A. Gallardo-Gutiérrez and J. R. Partington, *Multiplication by a finite Blaschke product on weighted Bergman spaces: commutant and reducing subspaces*, arXiv:2105.07760, especially PDF p. 3.
- [2] G. D. Taylor, *Multipliers on D_α* , Trans. Amer. Math. Soc. **123** (1966), 229–240, DOI: 10.1090/S0002-9947-1966-0206696-6.
- [3] D. A. Stegenga, *Multipliers of the Dirichlet space*, Illinois J. Math. **24** (1980), 113–139.
- [4] R. Kerman and E. Sawyer, *Carleson measures and multipliers of Dirichlet-type spaces*, Trans. Amer. Math. Soc. **309** (1988), 87–98, DOI: 10.1090/S0002-9947-1988-0957062-1.
- [5] D. Jupiter and D. Redett, *Multipliers on Dirichlet type spaces*, Acta Sci. Math. (Szeged) **72** (2006), 179–203; arXiv:math/0510193.
- [6] G. Bao, Z. Lou, R. Qian, and H. Wulan, *On multipliers of Dirichlet type spaces*, Complex Anal. Oper. Theory **9** (2015), 1701–1732, DOI: 10.1007/s11785-015-0444-0.